

VANCOUVER INTL, BC, Canada

WMO: 718920

Lat: 49.190N

Lon: 123.180W

Elev: 4

StdP: 101.27

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-5.1	-3.0	-12.8	1.2	-2.9	-9.1	1.7	-1.7	13.5	6.7	11.7	7.1	3.0	100	0.562

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.7	25.1	18.5	23.7	17.9	22.5	17.3	19.2	23.8	18.5	22.8	17.7	21.8	3.7	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	12.3	21.8	16.6	11.8	21.0	15.9	11.3	20.2	54.6	23.9	52.3	23.0	50.0	21.9	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
10.7	9.2	8.0	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-8.1	28.2	2.6	1.7	-9.9	29.5	-11.5	30.5	-12.9	31.4	-14.8	32.7
			-9.2	20.7	2.7	1.0	-11.1	21.4	-12.7	22.0	-14.3	22.6	-16.3	23.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.6	4.4	4.8	6.8	9.4	13.0	15.7	18.2	18.2	15.2	10.4	6.6	3.9
	DBStd	5.75	3.39	2.86	2.47	2.16	2.40	2.19	2.00	1.70	2.09	2.31	3.19	3.27
	HDD10.0	780	175	146	101	35	3	0	0	0	23	108	189	
	HDD18.3	2873	432	379	357	267	165	83	25	23	97	246	352	447
	CDD10.0	998	1	1	3	18	97	172	256	255	155	36	5	1
	CDD18.3	49	0	0	0	0	1	5	22	19	2	0	0	0
	CDH23.3	163	0	0	0	0	3	21	79	54	7	0	0	0
	CDH26.7	14	0	0	0	0	1	1	9	3	0	0	0	0
(14) Wind	WSAvg	3.7	3.6	3.6	4.1	3.9	3.7	3.7	3.8	3.4	3.4	3.5	3.8	3.8
(15) Precipitation	PrecAvg	1209	159	126	113	76	65	48	39	41	63	121	174	185
	PrecMax	1518	274	286	197	145	147	138	87	106	138	295	358	313
	PrecMin	843	28	34	48	16	23	14	0	0	0	21	49	56
	PrecStd	185	61	60	34	34	30	30	27	29	35	64	67	56
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	12.2	15.0	18.4	22.9	25.7	27.6	26.7	24.4	17.8	14.7	12.2
		MCWB	10.4	9.6	10.7	13.6	16.3	18.4	20.2	18.8	17.7	14.5	12.5	10.7
	2%	DB	10.8	10.7	12.8	16.2	20.3	23.3	25.3	24.8	21.8	16.0	13.0	10.2
		MCWB	9.5	8.6	9.8	12.0	15.0	17.3	18.9	18.4	16.7	13.3	11.5	8.9
	5%	DB	9.7	9.6	11.6	14.5	18.8	21.5	23.7	23.5	20.3	14.9	11.9	9.1
		MCWB	8.5	7.8	9.1	10.8	14.4	16.3	18.0	18.0	16.1	12.4	10.6	7.9
	10%	DB	8.8	8.6	10.5	13.2	17.4	20.1	22.5	22.2	19.0	13.9	10.9	8.1
		MCWB	7.6	7.0	8.3	10.0	13.5	15.4	17.3	17.5	15.3	11.9	9.7	7.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.1	10.5	11.6	14.2	16.8	19.1	20.9	19.9	18.8	15.4	13.1	11.0
		MCDB	11.7	11.5	13.3	17.8	21.6	24.2	27.0	24.3	22.4	17.0	13.9	12.0
	2%	WB	9.7	9.0	10.4	12.5	15.6	17.8	19.5	19.1	17.4	14.0	11.7	9.2
		MCDB	10.4	10.3	12.2	15.4	19.3	22.4	24.4	23.5	20.6	15.3	12.7	10.1
	5%	WB	8.6	8.0	9.4	11.4	14.8	16.8	18.5	18.5	16.6	12.9	10.8	8.1
		MCDB	9.5	9.2	11.2	13.9	18.2	20.9	23.1	22.7	19.5	14.4	11.7	8.9
	10%	WB	7.7	7.3	8.6	10.5	13.8	15.8	17.7	17.8	15.7	12.1	9.8	7.1
		MCDB	8.6	8.4	10.3	12.7	17.2	19.5	21.9	21.7	18.5	13.6	10.7	8.0
(35) Mean Daily Temperature Range	5% DB	MDBR	4.6	5.5	5.9	6.6	7.0	7.0	7.7	7.6	6.7	5.5	5.1	4.6
		MCDBR	4.8	5.9	6.9	8.4	9.1	9.4	9.5	9.3	8.7	6.3	5.2	4.6
		MCWBR	3.8	4.2	4.7	5.5	5.5	5.2	4.9	4.8	4.9	4.2	4.1	3.7
	5% WB	MCDBR	4.5	5.2	6.4	7.8	8.6	9.1	9.1	8.7	8.0	5.6	4.8	4.4
		MCWBR	3.7	3.9	4.7	5.4	5.4	5.3	4.9	4.8	4.8	4.0	4.0	3.7
(40) Clear-Sky Solar Irradiance	taub	0.320	0.314	0.331	0.363	0.387	0.363	0.359	0.363	0.346	0.337	0.322	0.308	
	taud	2.444	2.506	2.477	2.397	2.361	2.451	2.485	2.489	2.537	2.544	2.503	2.459	
	Ebn at Noon	736	845	885	885	873	897	895	876	859	805	731	699	
	Edh at Noon	64	75	90	108	117	108	103	98	85	72	60	55	
(44) All-Sky Solar Radiation	RadAvg	0.86	1.72	2.58	4.01	5.25	5.63	6.30	5.34	3.70	2.03	1.01	0.69	
	RadStd	0.14	0.26	0.35	0.43	0.46	0.67	0.55	0.42	0.33	0.28	0.19	0.10	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (67 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page